



**2025** CDISC + TMF  
EUROPE INTERCHANGE

**GENEVA**

CONFERENCE & EXPO: 14-15 MAY | TRAININGS: 12, 13, 16 MAY

## **ICH M11, TransCelerate, CDISC & HL7 Vulcan: Making the Electronic Protocol a Reality**

Dave Iberson-Hurst, USDM Product Owner & Technical Expert  
Version 2



# Meet the Speaker

Dave Iberson-Hurst

**Title:** CDISC USDM Product Owner

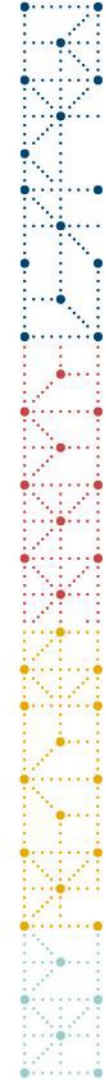
**Organization:** CDISC & data4knowledge

40 years' experience across several industries with the last 25+ years spent in the pharmaceutical industry combining his technology and software development experience with clinical trial process and data standards knowledge.

During this time, he has served as the CDISC CTO, worked on, and led, several CDISC teams, presented in many forums in Europe, the US, and elsewhere across the globe. He has worked closely with the FDA, EMA, HL7, ISO, and other standards organizations and was a member of CDISC's Blue Ribbon commission.

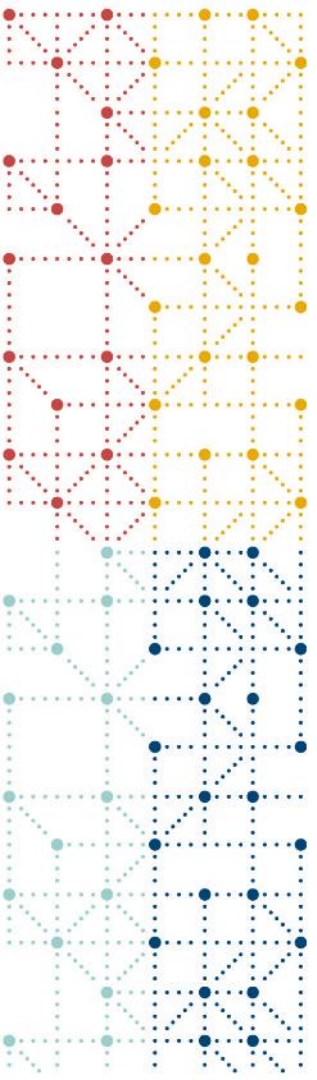
He is currently the CDISC Product Owner and Technical Expert for the Digital Data Flow project.

He is a partner at data4knowledge in Copenhagen and is focused on getting greater value and utility from clinical trial data.



# Disclaimer and Disclosures

- *The views and opinions expressed in this presentation are those of the author(s) and do not necessarily reflect the official policy or position of CDISC.*
- *On contract to CDISC for the DDF work*

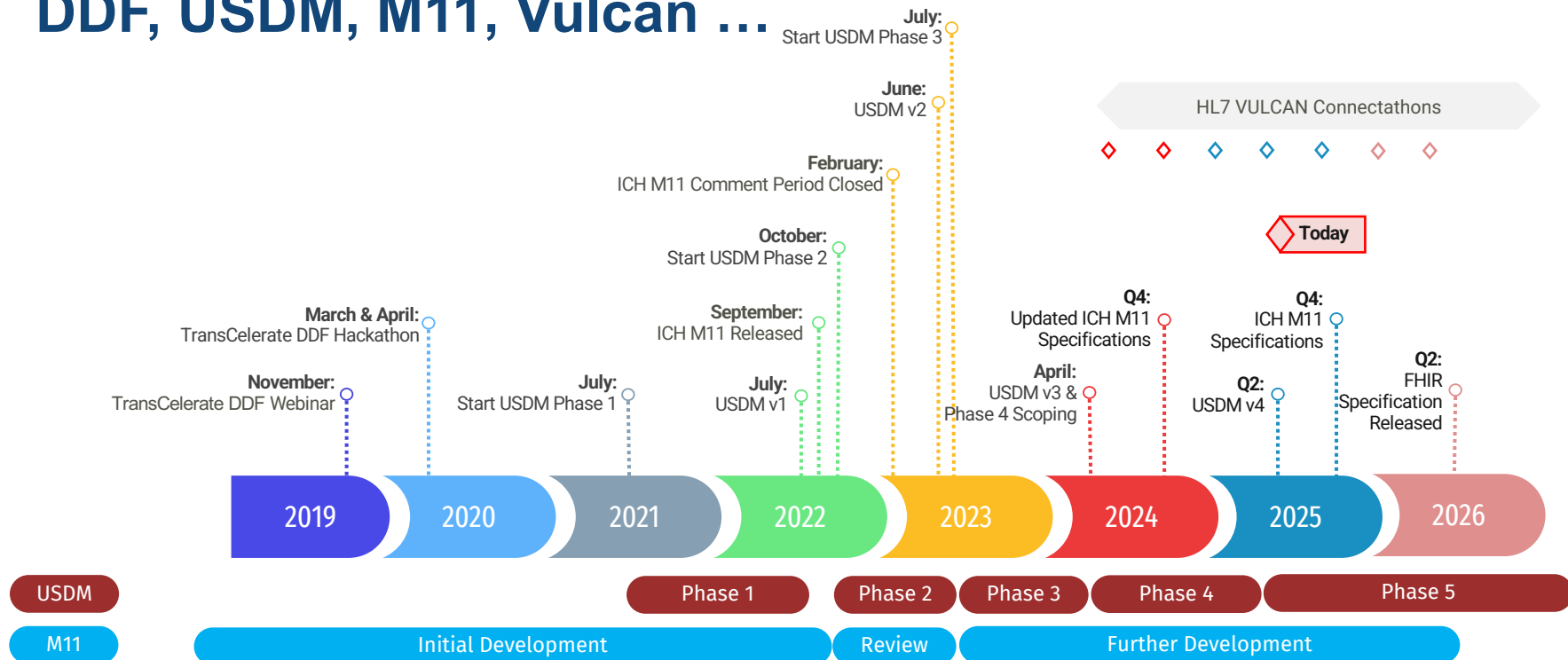


# Agenda

1. Digital Data Flow Project and USDM
2. ICH M11
3. USDM, M11, and HL7 UDP
4. precisionFDA and PRISM
5. And next ...



# DDF, USDM, M11, Vulcan ...



June 2018

## Acronyms

**DDF:** Digital Data Flow

**USDM:** Unified Study Definitions Model

**ICH:** International Council for Harmonisation

**M11:** Clinical electronic Structured Harmonised Protocol (CeSharP)

October 2022



## DDF 1

The first step in a journey. The base model providing an initial capability. Industry already pushing the boundaries of the model and using for a varied use cases



## DDF 2

DDF 2 just starting. Focused on two major use cases: EDC and CPT but the model will be expanded in other areas.



## Tipping Point?

When does the community recognise the benefits of an "electronic" study design / protocol? The "what is in it for me" question. There is a change management issue.



## Short Term Gains

Where is the win? The tangible, short term, gains ... CTMS, CT registries (e.g. CT.gov) and others

An eco-system of tools, APIs available off-the-shelf (the protocol API, the CTMS API ...) supported by well-understood model(s).

Implications for the CDISC products driving the need for "integrated", "consistent" and "aligned" standards



## Longer Term Challenge

2022

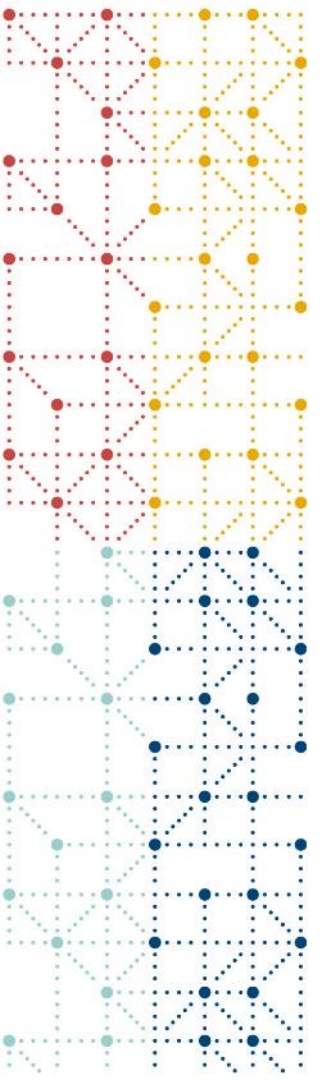
2023

2024

2025 +

Timeline indicative, NOT definitive

Time

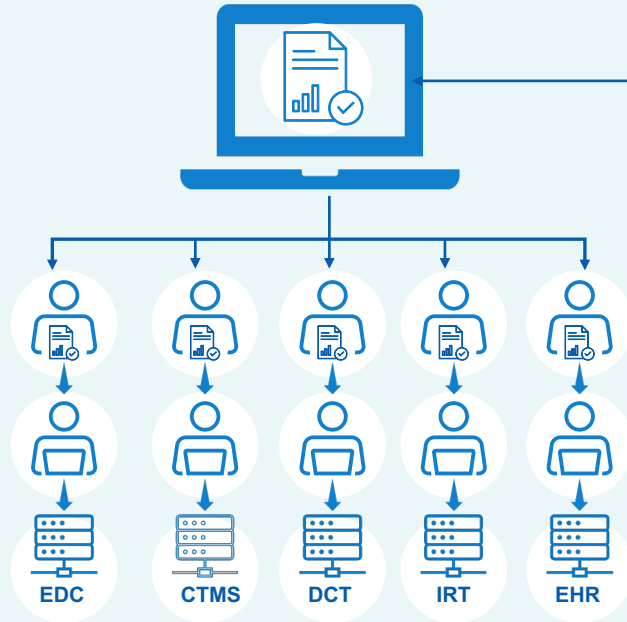


# Unified Study Definitions Model (USDm)

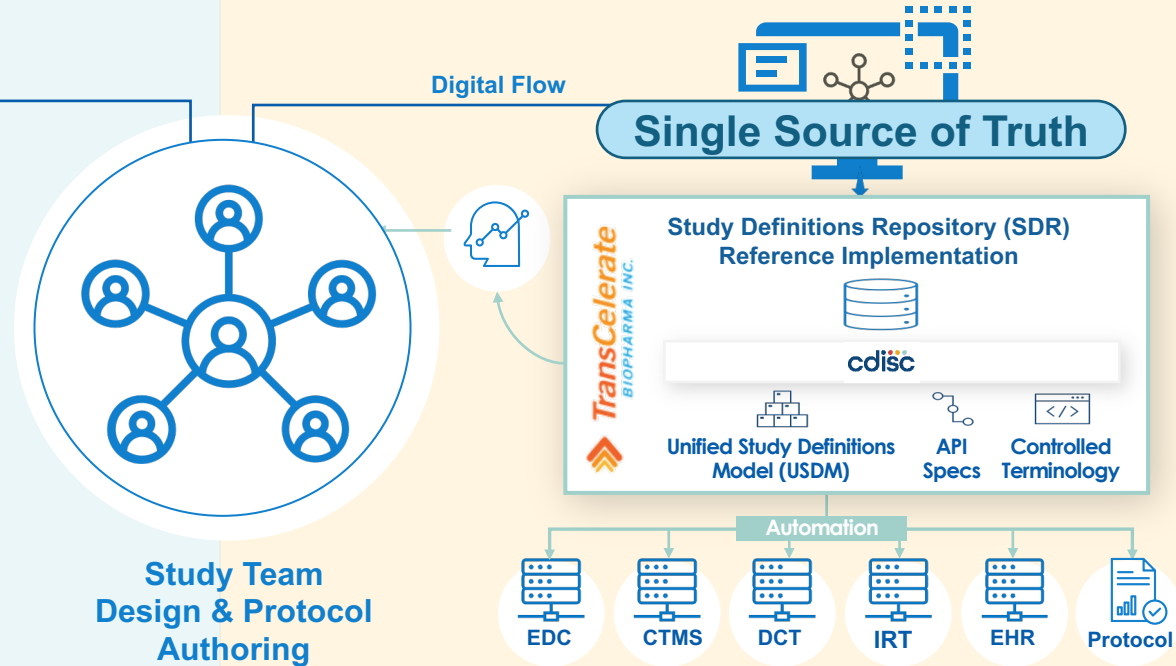
# TransCelerate Digital Data Flow (DDF) Ambition

*Write Once, Read Many*

**TODAY:** Document-based paradigm for protocol creation, interpretation, and transcription into consuming systems



**TOMORROW:** Digital paradigm for protocol creation, with fully automated data flow and interoperability between systems



# The USDM Standard

## CDISC Controlled Terminology

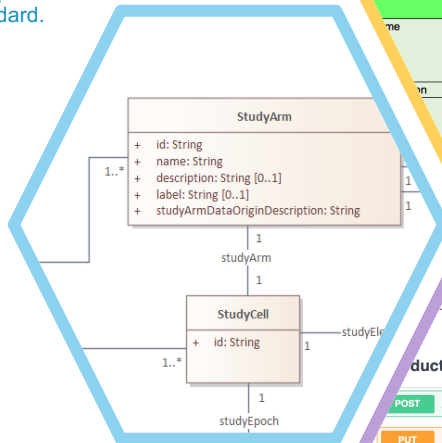
Provides further semantics, complementing the UML model.  
Includes the definition of classes, attributes, and value sets

## Examples

Example protocols implemented in the USDM with associated JSON files and visualisations

## Logical Model

The UML logical model (a class diagram) that provides the basis for the USDM standard.



## API Specification

Provides the means to exchange a single study between machines using a JSON API

## API for DDF

| Generate Digital Data Flow (DDF) Study Definitions Repository API. |                                                                  |
|--------------------------------------------------------------------|------------------------------------------------------------------|
| Introduction Routes that form the production specification.        |                                                                  |
| POST                                                               | /v3/studyDefinitions Create a study                              |
| PUT                                                                | /v3/studyDefinitions/{studyId} Update a study                    |
| GET                                                                | /v3/studyDefinitions/{studyId} Return a study                    |
| GET                                                                | /v3/studyDefinitions/{studyId}/history Returns the study history |
| POST                                                               | /v3/studyDesigns Study designs for a study                       |

## CORE Rules

Specification of the rules that define USDM compliance

| A                                                                                                                                                                        | B              | Entry Applies |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|
| Rule                                                                                                                                                                     | Warning/ Error | Entry Applies |
| Rule must conform with the USDM schema based on the schema based on the API specification.                                                                               | ERROR          | All           |
| Attributes (string, number, boolean) must conform with the schema based on the API specification.                                                                        | ERROR          | All           |
| Attributes must be included as defined in the USDM schema based on the API specification (i.e., all required properties are present and no attributes are present).      | ERROR          | All           |
| Attributes must be as defined in the USDM schema based on the API specification (i.e., required properties have at least one value and single properties are not lists). | ERROR          | All           |
| Within a study version, all id values must be unique.                                                                                                                    | ERROR          | All           |
| The names of all child instances of the same parent class must be unique.                                                                                                | ERROR          | All           |
| The same Biomedical Concept Category must not be referenced more than once from the same activity.                                                                       | ERROR          | Activity      |
| A specified biomedical concept category is expected to be referenced by an activity.                                                                                     | WARNING        | Activity      |
| A specified biomedical concept surrogate is expected to be referenced by an activity.                                                                                    | WARNING        | Activity      |
| Children must not refer to a timeline, procedure, concept, biomedical concept category or biomedical concept.                                                            | ERROR          | Activity      |
| A procedure is expected to be referenced by an activity.                                                                                                                 | WARNING        | Activity      |
| A timeline is expected to be referenced by an activity.                                                                                                                  | WARNING        | Activity      |
| Using the previous and next attributes must be used to reference children's concept category.                                                                            | WARNING        | Activity      |

## Implementation Guide

Guidance on using the USDM model and ensuring conformance with the standard

```
{
  "studyArms": [
    {
      "id": "StudyArm_1",
      "name": "Placebo",
      "label": "",
      "description": "Placebo",
      "type": {
        "id": "Code_61",
        "code": "C174268",
        "codeSystem": "http://www.cdisc.org",
        "codeSystemVersion": "2022-12-16",
        "decode": "Placebo Comparator Arm"
      }
    },
    {
      "studyArmDataOriginDescription": "Data collected from the study",
      "dataOriginType": {
        "id": "Code_62",
        "code": "C188866",
        "codeSystem": "http://www.cdisc.org",
        "codeSystemVersion": "2022-12-16",
        "decode": "Data Generated Within Study"
      }
    }
  ],
  {
    "id": "StudyArm_2",
    "name": "Xanomeline Low Dose",
    "label": "",
    "description": "Active Substance",
    "type": {
      "id": "Code_63",
      "code": "C174267",
      "codeSystem": "http://www.cdisc.org",
      "codeSystemVersion": "2022-12-16",
      "decode": "Active Comparator Arm"
    }
  }
]
```

on 2.0 Draft for Internal Review)



## Unified Study Definitions Model Implementation Guide (USDM-IG)

Version 2.0 (Draft for Internal Review)

Prepared by the DDF Team

Notes to Readers

- This is the draft version 2.0 of the Unified Study Definitions Model Implementation Guide (USDM-IG v2.0). It is intended for internal review only and is not a final version.

History

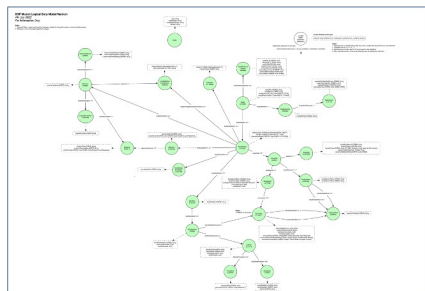
Version

2.0 Draft for Internal Review

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# CDISC DDF / USDM, Phases One to Four

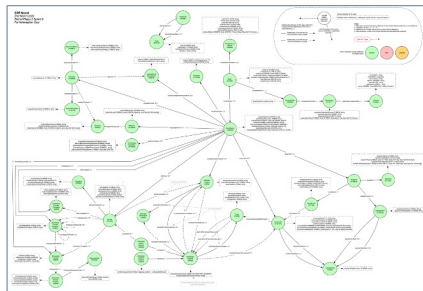
## Phase One



### 25 Classes

- Solid foundation
- The protocol document was an external entity into which the structured content could be exported

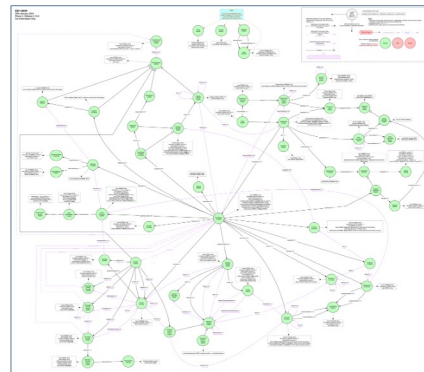
## Phase Two



### 35 Classes

- Focused on the structured elements of the protocol, e.g. the Schedule of Activities (SoA) & Biomedical Concepts (BCs)
- The protocol document still an external entity

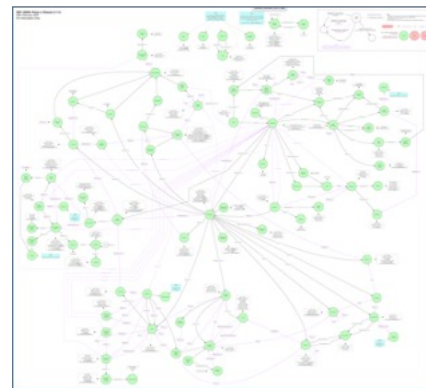
## Phase Three



### 58 Classes

- Now contains structured and unstructured elements
- The entire protocol document can be held within the USDM
- Allows for the protocol document to be generated from the model

## Phase Four

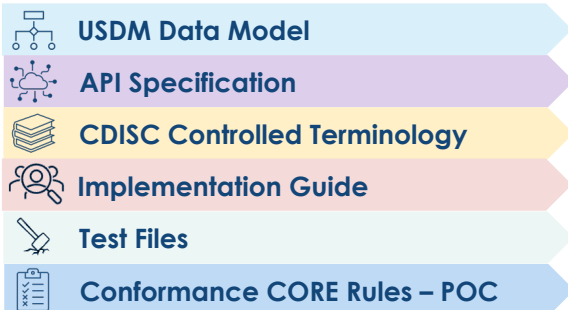


### 89 Classes

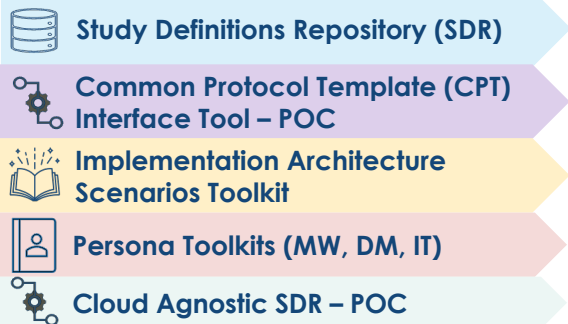
- Aligned with ICH M11
- Support for complex studies, interventional & observational studies, and medical devices
- Maximise content re-use and support for multiple document templates
- Extension mechanism to provide flexibility

# DDF Evolution: Phases One to Four

## CDISC's USDM Reference Architecture



## TransCelerate's SDR & Implementation Support



### PHASE ONE

July 2021 –  
July 2022



### PHASE TWO

Oct 2022 –  
Sep 2023



### PHASE THREE

July 2023–  
May 2024



### PHASE FOUR

Apr 2024–  
1Q 2025



TBD





# DDF Web Page

<https://www.cdisc.org/ddf>

[About](#)[Standards](#)[Tools](#)[Partnerships](#)[Education](#)[Events](#)[Membership](#)

[Home](#) / [Digital Data Flow](#)

## Digital Data Flow

[Overview](#)[Phase 4 Public Review](#)[What is the USDM](#)[Participate](#)[Webinars](#)[Versions](#)[FAQ](#)[Contact Us](#)



### Welcome to Digital Data Flow (DDF) for Clinical Trial Protocols

Digital Data Flow Initiative will help modernize clinical trials by enabling a digital workflow with protocol digitization. This initiative establishes a foundation for a future state of automated & dynamic readiness that can transform the drug development process.

Below are a list of the different websites sourcing specific content and resources. Depending on where you are in the journey, please feel free to explore the different websites and their information.



#### CDISC DDF Website

*You are here!*

Learn about and access the Unified Study Definitions Model (USDM) Reference Architecture supporting Digital Protocol Data



#### DDF Website

As the main website for the DDF initiative, learn and access all resources for DDF solutions



#### DDF GitHub Repos

Learn about a technical Reference Implementation of the USDM, the Study Definitions Repository (SDR), and access the



#### TransCelerate DDF Initiative Solutions

Learn about DDF Initiative background and roadmap

CDISC DDF



Scan Me



# DDF-RA (USDM) GitHub

<https://github.com/cdisc-org/DDF-RA>

The main screenshot shows the GitHub repository for **DDF-RA** (Public), generated from [cdisc-org/COSAHackathonTemplate](#). The repository has 49 issues, 11 pull requests, 1 Zenhub, 1 discussion, 1 action, 1 project, 1 wiki, 1 security, and 1 insight. The file structure is as follows:

| File                      | Description                                    | Last commit  |
|---------------------------|------------------------------------------------|--------------|
| <b>github-actions</b>     | Generate Diff and Data Dict                    | 6 months ago |
| <b>.github/workflows</b>  | use gh output instead of deprecated set-output | 6 months ago |
| <b>Deliverables</b>       | Generate Diff and Data Dict                    | 2 weeks ago  |
| <b>Documents</b>          | Update informative diagram for criteria tweak  | 3 weeks ago  |
| <b>HowTos</b>             | Initial commit                                 | 3 weeks ago  |
| <b>images</b>             | Initial commit                                 | 3 weeks ago  |
| <b>.gitignore</b>         | Updated Objective and Endpoint                 | 3 weeks ago  |
| <b>CODE_OF_CONDUCT.md</b> | Updated the enforcement contact                | 3 weeks ago  |
| <b>CONTRIBUTING.md</b>    | Initial commit                                 | 3 weeks ago  |
| <b>LICENSE</b>            | Update copyright holder                        | 3 weeks ago  |
| <b>README.md</b>          | Update README.md                               | 3 weeks ago  |

The inset shows the **DDF-RA / Deliverables** directory view, listing the following files and their last commit messages:

| Name  | Last commit message               |
|-------|-----------------------------------|
| ..    |                                   |
| API   | API updates                       |
| CT    | Update USDM_CT.xlsx               |
| IG    | Added USDMIG post copy editing    |
| RULES | Set of USDM v3.0 rules            |
| UML   | Pushed to wrong branch, reverting |

CDISC GitHub



Scan Me

# USDM in Action

## Use Cases Supporting the DDF Vision

### Retrospective

### Use Cases

### Prospective

#### Protocol 'Store'

##### Past Protocols

The storage of past sponsor protocols in to support a variety of use cases such as:

- Therapeutic area specific content
- Standard inclusion and exclusion criteria.
- Libraries of objectives, endpoints and estimands.
- Asses the past to prevent protocol amendments.
- General search across the library

#### Study Design

##### Protocol Authoring

Support the authoring of new protocols including complex studies such as Master, Umbrella etc.

This would include analyzing previous protocol information to inform design choices (see Protocol Store).

Writing associated SAP, Informed Consent, Monitoring Plan, Drug Plan etc via content reuse.

##### Study Build

Build of the study based on the protocol including all detailed procedures and assessments to allow for immediate deployment.

##### Feasibility & Cost

Support study feasibility and complexity evaluations.

Determine the cost of the study including such items as vendor and CRO costs and support the laboratory RFP process

##### Regulatory Authority

Approval for the execution of the study including IND, CTA etc and discussions with regulatory authorities

##### Stakeholder Views

Tailored views for stakeholders such as IRB and ethic committees.

#### Study Start-up

##### Deployment

Provide a solid foundation for the study across the USDM to support electronic configuration rather than manual data entry including Data collection systems (Incl. EDC), CTMS, IRT, Data transmission specifications and more

##### TMF

Linking the protocol to the specification of milestones and essential documents

##### Data Collection Strategy

Ensure all data that is needed for analysis is collected. Remove unnecessary procedures & data collection to reduce trial cost and patient burden.

##### Trial Registries

Registration of the study with registries such as CT.gov, CTR etc.

##### Training

Provision of training materials to those involved in the study including sites

##### Site Support

Provision of tailored information to sites regarding study execution.

#### Study Execution

##### Amendments

Support the amendment process with the ability to indicate the precise changes to the user community.

Ensure fewer amendments (see Protocol Store).

##### Subject Information

Provision of improved information to subjects regarding study execution, improve expectation management.

##### Drug & Study Materials

Assist in the planning and provision of study drug, materials and supplies

##### Subject Recruitment

Support the identification of subjects to help improve subject recruitment to the study. Help support EHR participation identification.

#### Analysis & Reporting

##### Automation

Provide a solid foundation for the automation of analysis and reporting outputs such as TLFs and CSRs.

##### SDTM

Creation of 'T' domains. Auto generation of data domains

##### ADaM

Use the protocol statistical metadata to assist in the derivation of analysis datasets

##### Statistical Analysis Plan

Use the protocol as the initial source of the SAP to ensure consistency with objectives, endpoints, estimands

#### Regulatory Submission

##### Submission

Preparation of the submission by the sponsor

##### Regulatory Review

Review of the submission by the regulatory authority and discussions / questions with the sponsor

##### Statistical and Safety Review

Support the comparison of data by arm or study design.

Allow for the verification of planned (protocol) data versus actual (study data) captured data

### Across The Lifecycle

#### Dashboards

The display of protocol related information sourced from one or more protocols in ways which have not been achievable before

#### Comparison

Compare and verify differences across versions and similar protocols

### Unified Study Definitions Model (USDM)

#### Controlled Terms

#### Unstructured Content

Populations, Inclusion & Exclusion, Interventions & Indications, Estimators, Objectives & Endpoints

Study, Identifiers, Amendments

Study Designs, Arms, Epochs

Procedures, Biomedical Concepts

Detailed Study Logic, Encounters

#### Study Details

The overall study, its various versions, identifiers and associated governance. Also includes the amendments made to the study

#### Study High Level Design

The single or set of study designs making up the study detailing the epochs, arms etc.

#### Study Science

The detailed description of the study science: the populations and the associated inclusion and exclusion criteria, the indications being studied, the interventions being used and the objectives, endpoints and the associated estimands.

#### Detailed Study Logic

A precise definition of the study logic including support for the Schedule of Activities.

#### Unstructured Content

The ability to support one or more document presentations of the USDM content including the ICH M11 protocol template, sponsor templates and other documents.

#### Procedures and Biomedical Concepts

The detail around the procedures and observations to be performed as part of the detailed study designs.

#### Controlled Terms

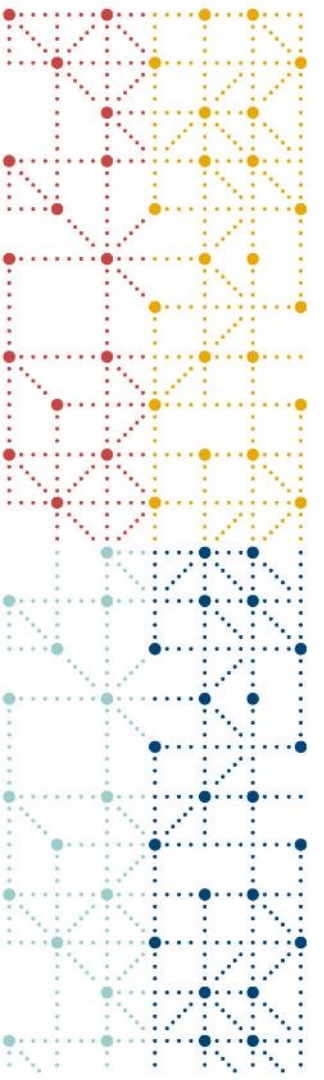
The controlled terminology needed to define the semantics within the model. Managed in the same manner as all CDISC CT and aligned with the M11 template standard.

### Use Cases



Scan Me

NOTE: The use cases presented are illustrative and the list is not intended to be exhaustive.

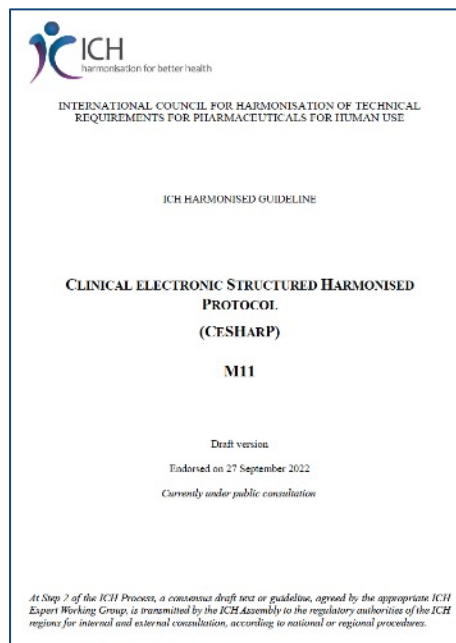


# ICH M11

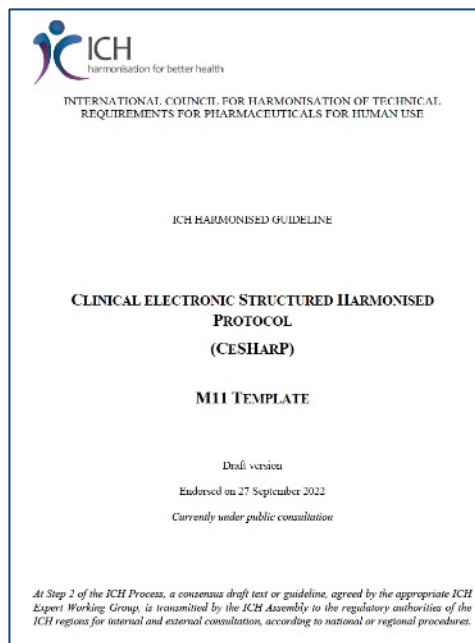
# M11 Is ...

## ICH CLINICAL ELECTRONIC STRUCTURED HARMONISED PROTOCOL (CeSHaRP)

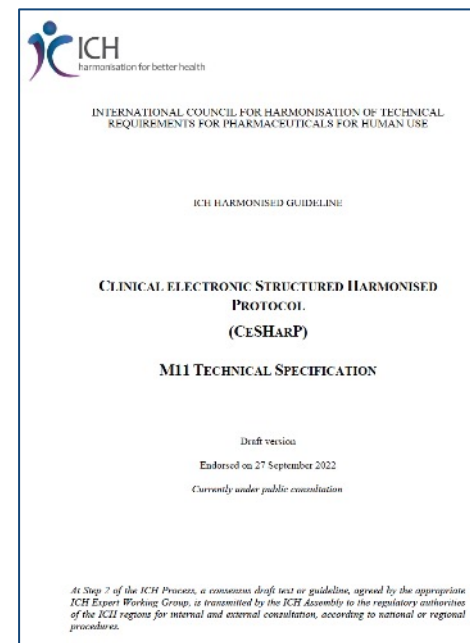
<https://www.ich.org/page/multidisciplinary-guidelines>



Provides background, purpose, and scope as a guideline



Provides the written format for the  
Interventional Clinical Trial Protocol  
Template

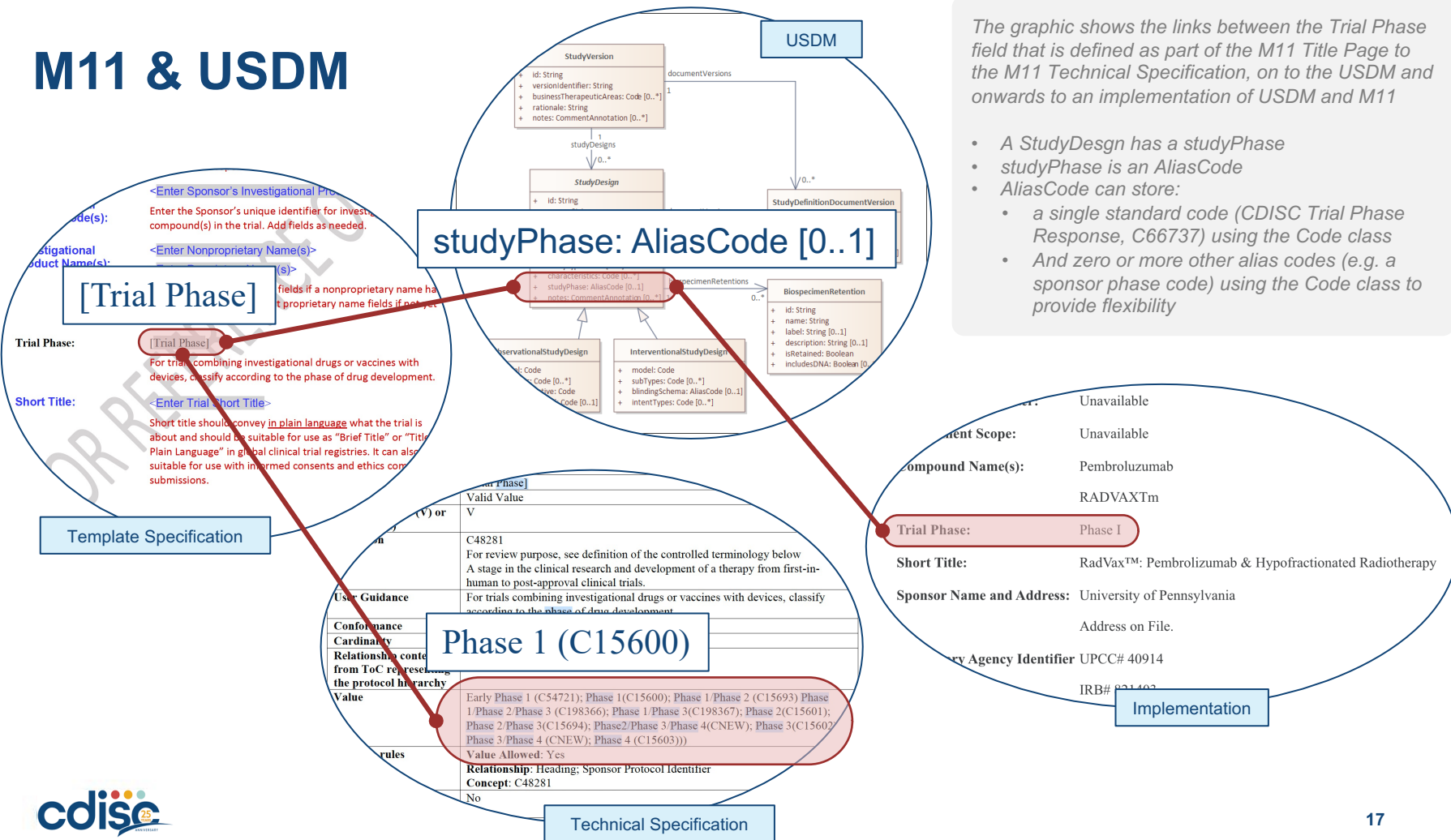


Provides the technical representation  
aligned with the guideline and protocol  
template

# M11 & USDM

The graphic shows the links between the Trial Phase field that is defined as part of the M11 Title Page to the M11 Technical Specification, on to the USDM and onwards to an implementation of USDM and M11

- A StudyDesign has a studyPhase
- studyPhase is an AliasCode
- AliasCode can store:
  - a single standard code (CDISC Trial Phase Response, C66737) using the Code class
  - And zero or more other alias codes (e.g. a sponsor phase code) using the Code class to provide flexibility





# ICH and CDISC MOU (Memorandum of Understanding)

As a collaboration between ICH and CDISC, the goals of the agreement are to:

- Use a unified governance process and terminology services for the long-term support of ICH controlled terminologies
- Curate and maintain ICH controlled terminologies
- Follow a robust process for the public review and publication of ICH terminologies
- Ensure the terminologies are freely available to the public following public review

## Scope

For ICH members to adopt and implement a clinical information standard it is critical that all terminology components, including but not limited to definitions described in the technical specification, are part of a greater international controlled terminology resource managed by an internationally recognized standards development organization (SDO). CDISC has been identified by ICH as a reputable SDO with the qualifications and capabilities to support the maintenance and facilitation of the governance process for ICH controlled terminology.

This Memorandum of Understanding (MOU) sets forth the roles and responsibilities of each party as they relate to the governance of the ICH terms and definitions developed in collaboration with CDISC. This MOU is intended to describe the goals, the high-level governance process, and how each party will collaborate. Specific projects (e.g., M11 controlled terminology) will be defined in detail as part of an annex to this MOU mutually agreed upon by CDISC and ICH.

## Goals

As a collaboration between ICH and CDISC, the goals of the agreement are to:

1. Use a unified governance process and terminology services for the long-term support of ICH controlled terminologies.
2. Curate and maintain ICH controlled terminologies.
3. Follow a robust process for the public review and publication of ICH terminologies
4. Ensure the terminologies are freely available to the public following public review.





# M11 and USDM Alignment

Created by: J. Benoit-Mont, d4knowledge ApS  
Version 1, 2023-03-28  
Based On  
M11: Updated Step 2 Draft 2021-03-19  
USDM: Version 3.12 2020-03-04

This infographic shows the relationship between the M11 protocol template specification and the structured and unstructured components of the USDM. Each highest area of the M11 protocol template is highlighted along with the associated area within the USDM.

The "mapping" is not precise. It is intended as a guide to show how the elements of a protocol document can be placed within the model and how the document can be generated from a populated model.

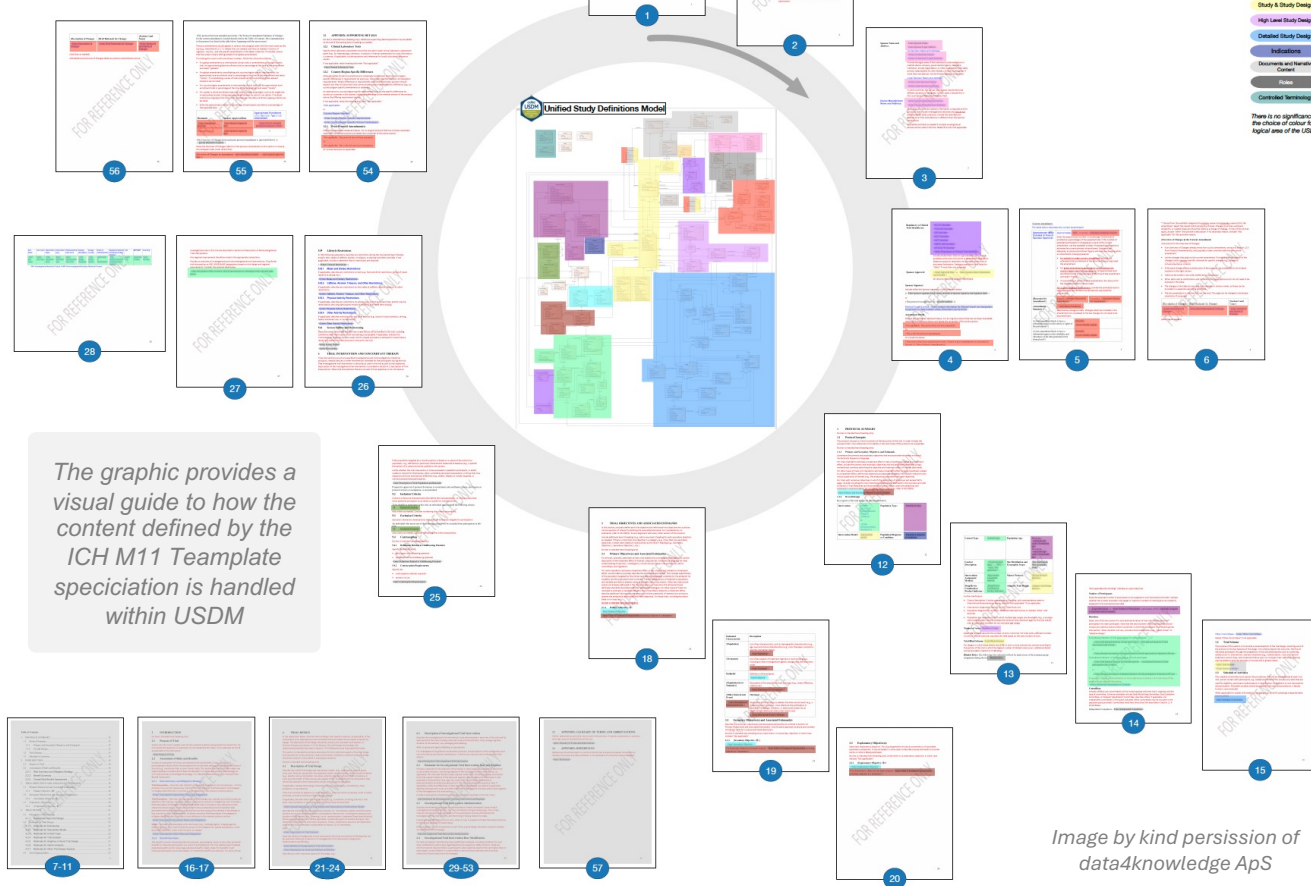
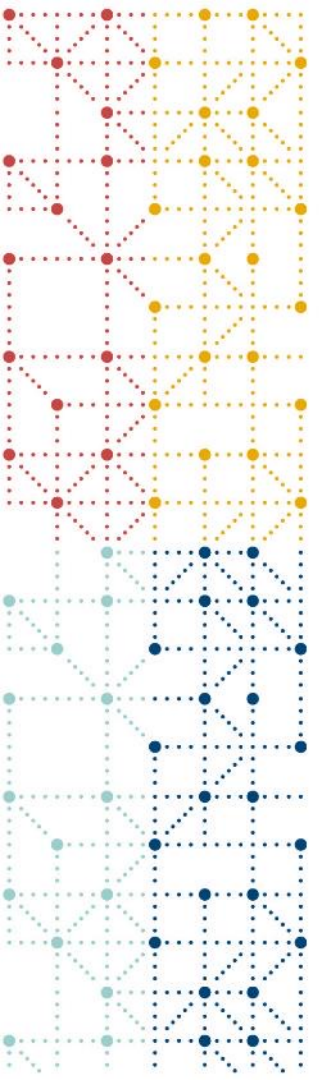


Image by kind persission of data4knowledge ApS

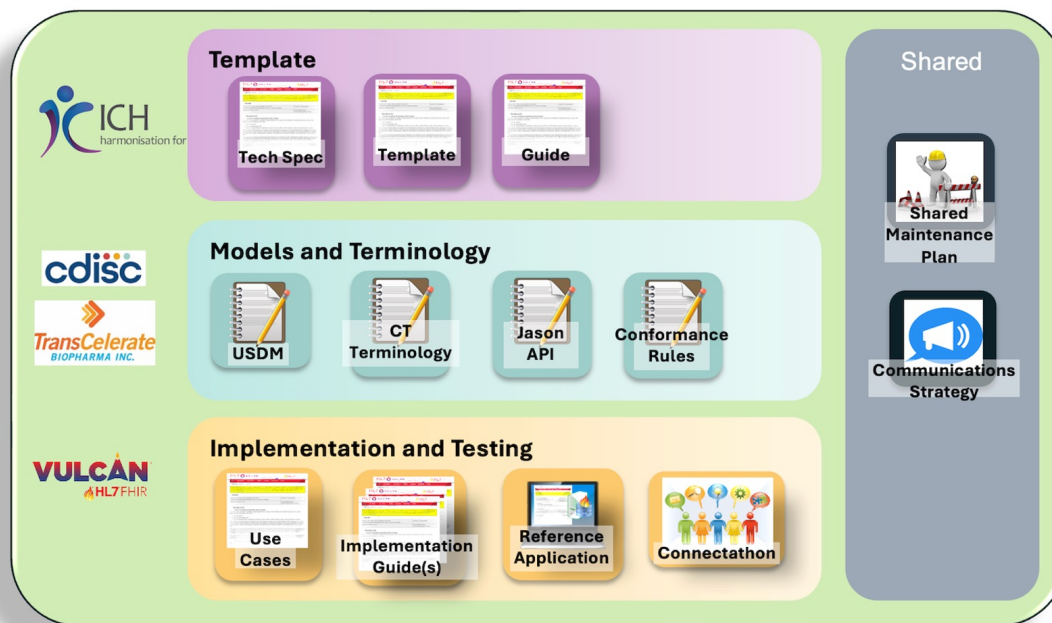


## USDM, M11, and the HL7 UDP



## Utilizing the Digital Protocol

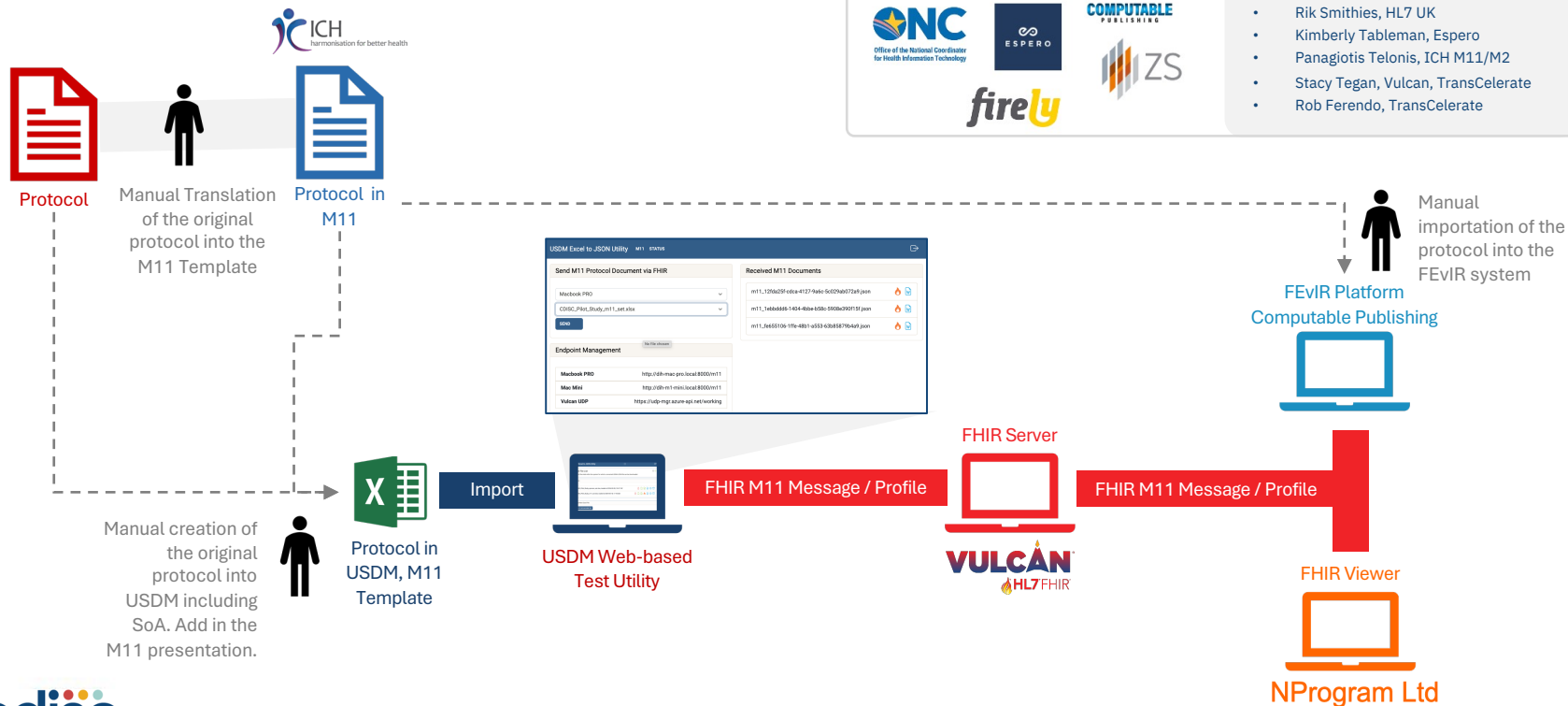
UDP is an Umbrella Project with many Use Cases



- **Inputs:**
  - ICH M11 template
  - ICH M11 technical specification
  - Models, definitions
- **FHIR will carry CDISC CT and USDM content**
- **The technical specification can be used to develop other Implementation Guides**

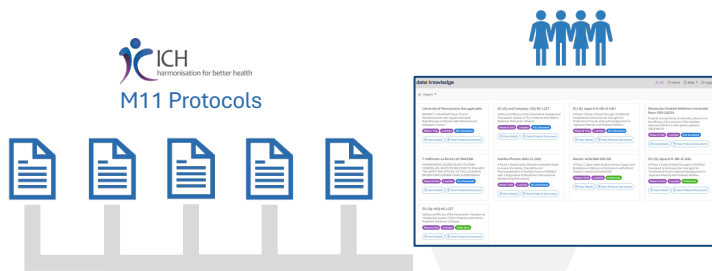
# HL7 FHIR Connectathon

Dallas, May 2024



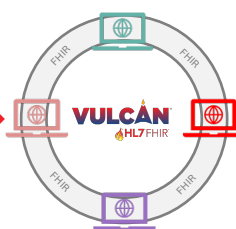
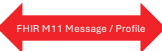
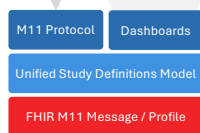
# HL7 FHIR Connectathon

Atlanta, September 2024



**“Sponsor”** 1

- M11 ‘.docx’ → USDM → FHIR

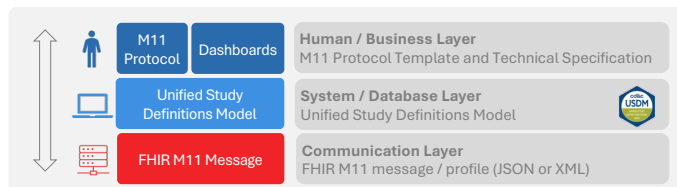


**“Regulator”** 2

- FHIR → USDM → M11 Presentation

Structure parts of the M11 Template:

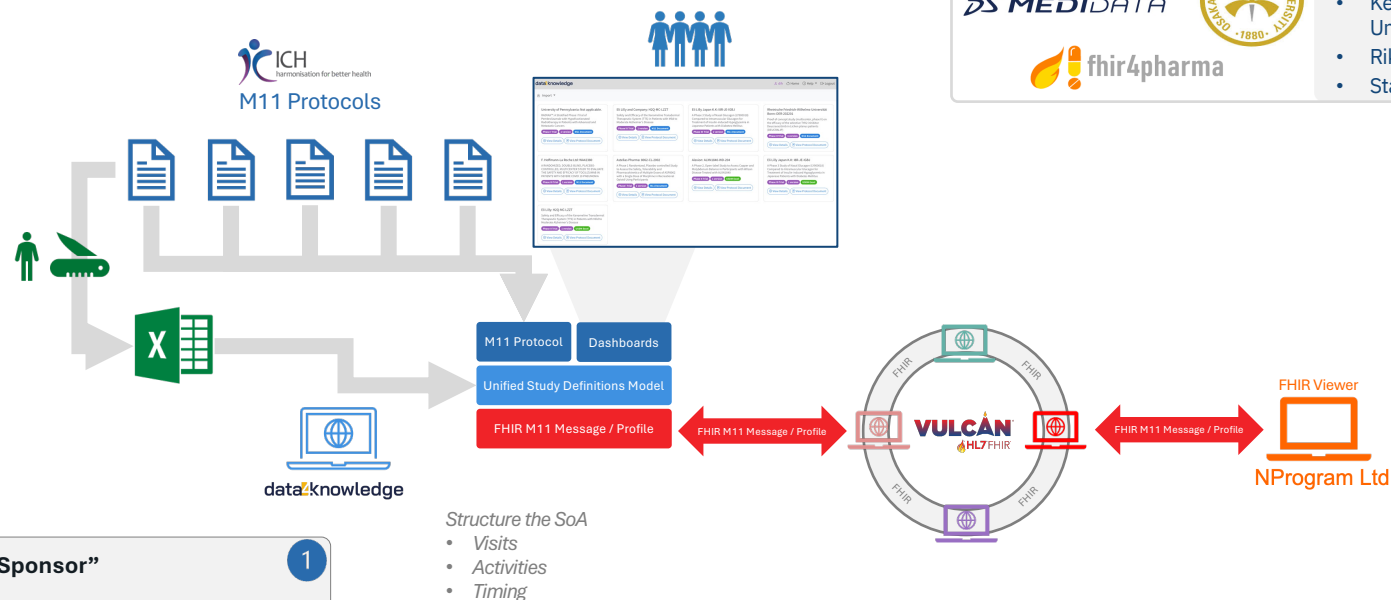
- Title Page
- Inclusion / Exclusion
- Estimands, Objectives and Endpoints
- Amendments



- Brian Alper, Computable Publishing
- Hugh Glover, Vulcan TRACK LEAD
- Dave Iberson-Hurst, CDISC
- Khalid Shahin, Computable Publishing
- Rik Smithies, HL7 UK
- Stacy Tegan, Vulcan, TransCelerate
- Rob Ferendo, TransCelerate
- Mary Lynn Mercado, Novartis
- Enam Rahman, TransCelerate
- Mihoko Okada, Idial



## Virtual, January 2025

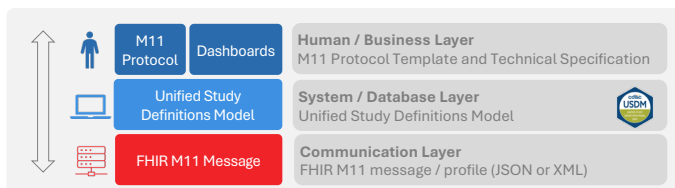


## “Sponsor”

- M11 '.docx' → USDM → FHIR M11
- USDM '.xlsx' → USDM → FHIR SoA

## Structure the SoA

- Visits
- Activities
- Timing



- Hugh Glover, Vulcan TRACK LEAD
- Rob Ferendo, TransCelerate
- Patrick Genyn, fhir4pharma
- Dave Iberson-Hurst, CDISC
- Geoff Low, Medidata
- Keiko Ota, Osaka Metropolitan University Hospital
- Rik Smithies, HL7 UK
- Stacy Tegan, TransCelerate, Vulcan

## “Regulator”

- FHIR → M11 Presentation

## “Operational Environment”

- FHIR → SoA Presentation



# HL7 Connectathon, Madrid, May 2025



## Vulcan HL7 Connectathon Highlights

Founded 6 years ago: supporting a membership of 50+ organisations and growing, Academia, Government, Consortia/SDO, Health Tech/Software Implementors, BioPharmaceutical companies

Scope: FHIR across **Translational Research** <-> **Clinical Research** <-> **Clinical Care**

UDP (Utilising the Digital Protocol) -> in Partnership with **cdisc** **TransCelerate** BIOPHARMA INC.

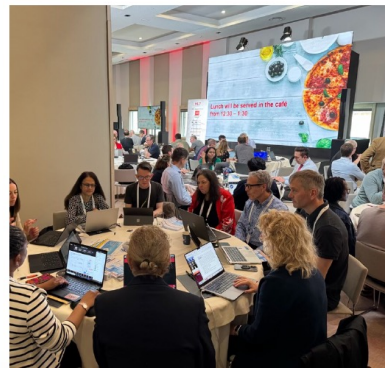
- Targeting the September Ballot
- very productive conversation with
  - ICH (International Council for Harmonization)
  - EBM (the FHIR Resources for Evidence Based Medicine) and UDP

IMI (eMedicinal-product-info) -> in Partnership with **Gravitate** **Health**

from narrative text to structured resources <-> and back  
assessment of the Medication Resources in FHIR  
/IDMP: increasing the resilience of the global mechanisms for patient safety, cross-border prescription & drug shortages by  
and publishing global identifiers for pharmaceutical products and substances using FHIR mechanisms.

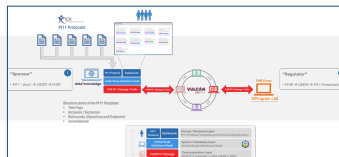
European Activities  
October 2025 in Copenhagen Partnering with **TEHDAS** (Towards the European Health Data Space)

IDMP Working Group  
Medication of Medicinal Products





## Madrid, May 2025



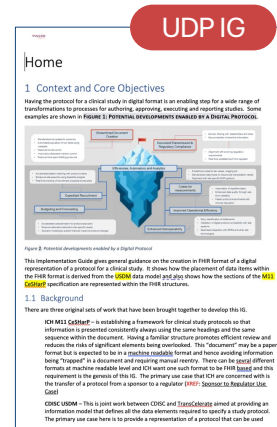
## Dallas & Atlanta Connectathons



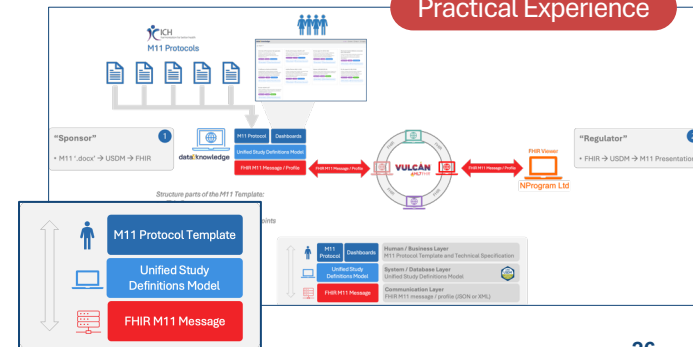
## ICH M11 Template & Technical Specifications

| Table 1. List of the 100 most cited articles in the literature on the topic of "The Role of the Teacher in the 21st Century".<br>Source: Scopus database, 2018. Data as of 15/05/2018. Last update: 15/05/2018. |      |            |                               |           |            | Keywords                                                                                                                                                                                                                                                                                                                         | Other references                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|-------------------------------|-----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rank                                                                                                                                                                                                            | Year | Authors    | Journal                       | Citations |            |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 1st                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. |
| 2nd                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 3rd                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 4th                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 5th                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 6th                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 7th                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 8th                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 9th                                                                                                                                                                                                             | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 10th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 11th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. |
| 12th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 13th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 14th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 15th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 16th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 17th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 18th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 19th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 20th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 21st                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. |
| 22nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 23rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 24th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 25th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 26th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 27th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 28th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 29th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 30th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 31st                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. |
| 32nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 33rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 34th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 35th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 36th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 37th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 38th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 39th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 40th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 41st                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. |
| 42nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 43rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 44th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 45th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 46th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 47th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 48th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 49th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 50th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 51st                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. |
| 52nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 53rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 54th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 55th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 56th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 57th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 58th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 59th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 60th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
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| 62nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 63rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 64th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 65th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 66th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 67th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 68th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 69th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 70th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
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| 72nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 73rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 74th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 75th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 76th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 77th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 78th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 79th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 80th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
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| 82nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 83rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 84th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 85th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 86th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 87th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 88th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 89th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 90th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 91st                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. | The article discusses the role of the teacher in the 21st century, focusing on the need for teachers to be able to adapt to a rapidly changing world. It argues that teachers must be able to work in a collaborative way, and that they must be able to use a variety of teaching strategies to meet the needs of all learners. |
| 92nd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 93rd                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 94th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 95th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 96th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 97th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 98th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 99th                                                                                                                                                                                                            | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| 100th                                                                                                                                                                                                           | 2006 | Fullan, M. | Journal of Curriculum Studies | 10,173    | Fullan, M. |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |

ICH M11 and CDISC  
USDM share the same  
CT

CDISC USDM  
Logical Model

## Practical Experience



# ICH M11 Work Plan

## Next ICH Assembly meetings

13 - 14 May 2025 in Madrid, Spain

18 - 19 November 2025 in Singapore

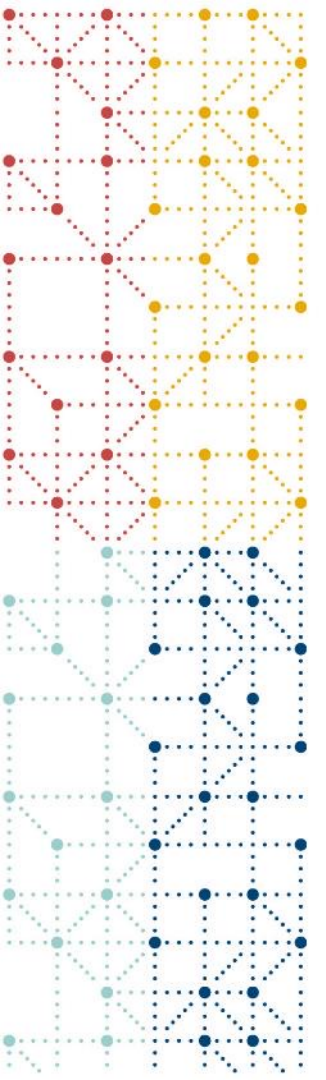
### 1.c. Future anticipated key milestones

| Expected future completion date | Milestone                                                                                                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Oct. 2024</b>                | <ul style="list-style-type: none"><li>Regional Party Review of the Updated Guideline, Template, and Technical Specification</li><li>Clinical Data Interchange Standards Consortium (CDISC) Public Review of the M11 terms, definitions, and valid values</li></ul> |
| <b>Dec 2024</b>                 | <ul style="list-style-type: none"><li>Adjudication of review comments</li><li>Updated Guideline, Template and Technical Specification</li></ul>                                                                                                                    |
| <b>Mar. 2025</b>                | Regional Public Consultation Period on Technical Specification                                                                                                                                                                                                     |
| <b>Jun 2025</b>                 | Adjudication of Public Comments on the Technical Specification                                                                                                                                                                                                     |
| <b>Sep 2025</b>                 | Updated Guideline, Template and Technical Specification                                                                                                                                                                                                            |
| <b>Oct. 2025</b>                | Step 3 Sign-off Guideline, Template and Technical Specification                                                                                                                                                                                                    |
| <b>Nov. 2025</b>                | Step 4 adoption of Guideline, Template and Technical Specification                                                                                                                                                                                                 |
| <b>Nov. 2025</b>                | Final versioned training materials                                                                                                                                                                                                                                 |
| <b>Feb 2026</b>                 | Step 2 (Testing) of the ICH Technical Implementation Guide for Fast Healthcare Interoperability Resources (FHIR)                                                                                                                                                   |
| <b>May 2026</b>                 | Step 4 adoption of ICH Technical Implementation Guide for FHIR                                                                                                                                                                                                     |

Today

Release

FHIR IG



## precisionFDA and PRISM



Sponsors



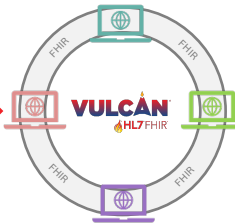
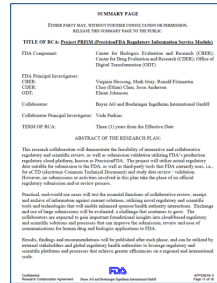
ICH  
M11 Protocols



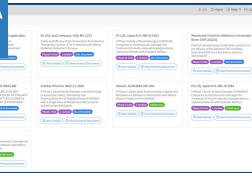
Sponsor

1

- M11 '.docx' → USDM → FHIR
- Try and make it easy, "a few clicks"



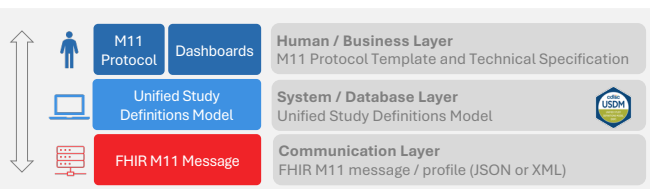
## precisionFDA Regulatory Information Service Module FDA-Industry Research Collaboration Agreement (Public-Private Partnership)

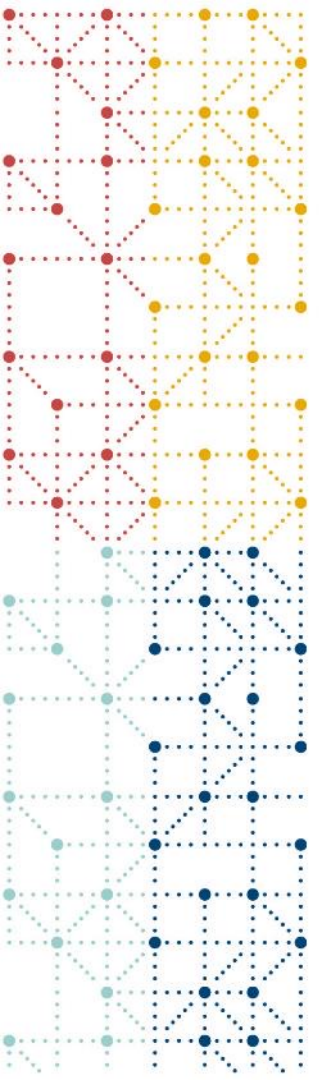


FDA

2

- FHIR → USDM → M11 Presentation
- Dashboards, what is possible?

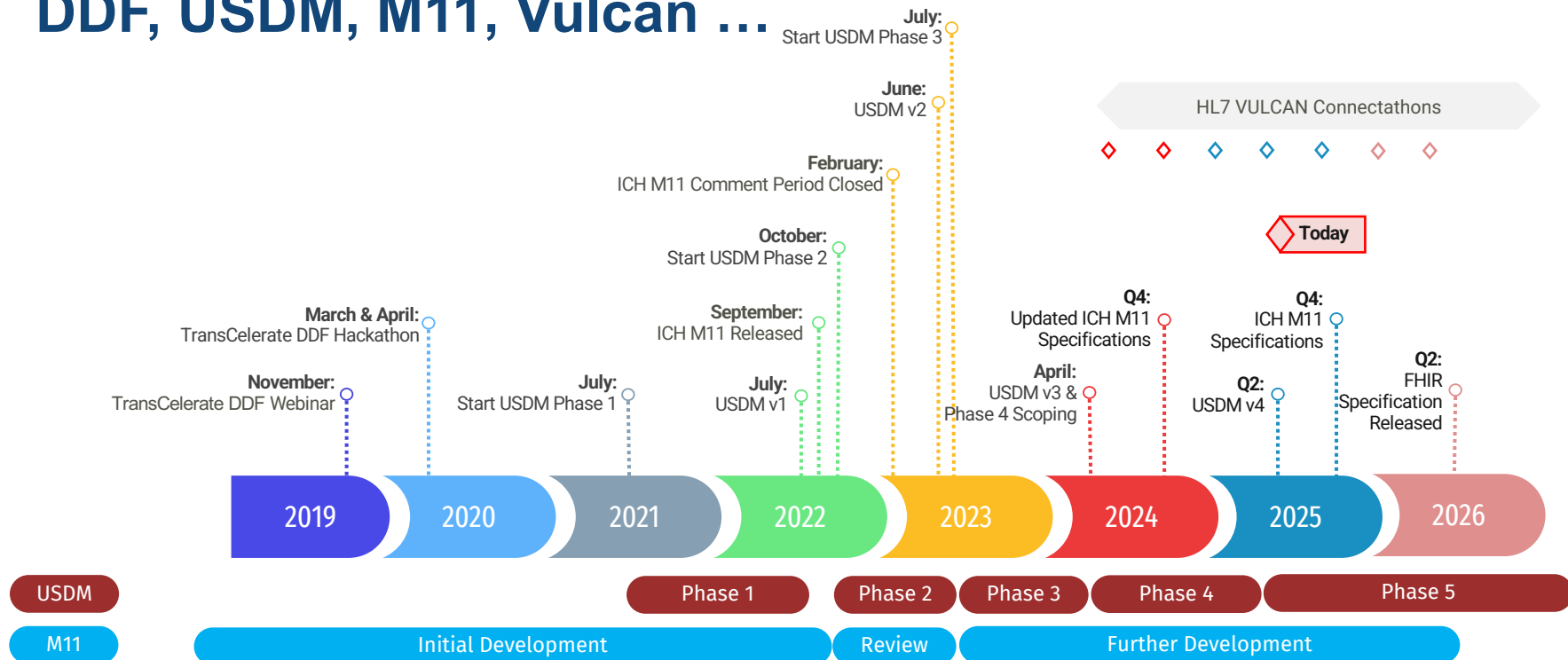




**And next ...**



# DDF, USDM, M11, Vulcan ...



June 2018

## Acronyms

**DDF:** Digital Data Flow

**USDM:** Unified Study Definitions Model

**ICH:** International Council for Harmonisation

**M11:** Clinical electronic Structured Harmonised Protocol (CeSharP)



ICH  
harmonisation for better health



**USDM Phase 5**  
Stability. The foundation for  
CDISC 360i. Scoping ongoing.



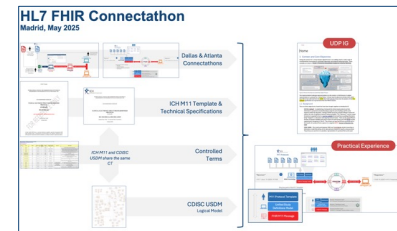
precisionFDA Regulatory Information Service Module  
FDA-Industry Research Collaboration Agreement  
(Public-Private Partnership)

**FDA & PRISM**  
Working with FDA to pilot first electronic transfer of an M11 protocol as well as tooling to support



**HL7 Vulcan & UDP**  
Working with HL7 Vulcan  
to build FHIR message to  
support exchange of  
USDM / M11 content.

**VULCAN~UDP**



### EMA, M11 & CTIS



## SAVE THE DATE

DDF: MISSION POSSIBLE

## PRACTICAL APPROACHES FOR PROTOCOL DIGITALIZATION

24-25  
SEPTEMBER

NOVARTIS IN NEW JERSEY, U.S.

**TransCelerate & Adoption**  
Sponsor and vendors working with USDm. Latest adoption will be visible at the TransCelerate 'Mission Possible' days in September



October 2022



## DDF 1

The first step in a journey. The base model providing an initial capability. Industry already pushing the boundaries of the model and using for a varied use cases



## DDF 2

DDF 2 just starting. Focused on two major use cases: EDC and CPT but the model will be expanded in other areas.



## Tipping Point?

When does the community recognise the benefits of an "electronic" study design / protocol? The "what is in it for me" question. There is a change management issue.



## Short Term Gains

Where is the win? The tangible, short term, gains ... CTMS, CT registries (e.g. CT.gov) and others

An eco-system of tools, APIs available off-the-shelf (the protocol API, the CTMS API ...) supported by well-understood model(s).

Implications for the CDISC products driving the need for "integrated", "consistent" and "aligned" standards



## Longer Term Challenge



2022

2023

2024

2025

2026

Timeline indicative, NOT definitive

## DDF in Action

Transforming Clinical Trials with Standards and Digitalization  
"Continuing the Journey, Charting the Future"

Norve Nordisk, Copenhagen  
10 October 2024

## SAVE THE DATE

DDF MISSION POSSIBLE  
PRACTICAL  
APPROACHES FOR  
PROTOCOL  
DIGITALIZATION  
24-25 SEPTEMBER  
2024  
AMSTERDAM, NETHERLANDS  
A CONFERENCE CO-LOCATED WITH CDISC

# Thank You!



*HL7 Connectathon, Atlanta, Sept 2024, College Football Hall of Fame*

## CDISC Team:

- Dave IH
- Richard Marshall
- Erin Muhlbradt
- John Owen
- Berber Snoeijer
- Craig Zwickl

## Contacts:

Dave IH: [diberson-hurst@cdisc.org](mailto:diberson-hurst@cdisc.org)

John Owen: [jowen@cdisc.org](mailto:jowen@cdisc.org)

J Owen Email



Scan Me

DIH Email



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